Exercise 10.1 SOLUTION

Q1. Without using the tables, find the values of

i. $\cos(-1230^\circ)$

Sol: $\cos(-1230^\circ) = \cos(1230^\circ)$

$= \cos(13\times 90+60)^\circ = -\sin 60^\circ = -\frac{\sqrt{3}}{2}$

نوٹ:

ان تمام سوالات میں جو بھی نمبر دیا گیا ہے تقسیم کر کے چیک کرنا ہے کہ اس میں پہلے کتنی دفعہ 90° منفی کریں تو باقی کیا بچتا ہے مثلاً جیسے 1230° میں سے 13 دفعہ 90° منفی کریں تو باقی 60° بچتا ہے اور لکھنا ایسے ہے

$1230^\circ = 13 \times 90^\circ + 60^\circ$

ii. $\tan(-1035^\circ)$

Sol: $\tan(-1035^\circ) = -\tan(1035^\circ)$

$= -\tan(11\times 90+45)^\circ = -(-\cot 45^\circ) = \cot 45^\circ = \frac{1}{\tan 45^\circ} = \frac{1}{1} = 1$

iii. $\sec(1140^\circ)$

Sol: $\sec(1140^\circ) = \sec(12\times 90+60)^\circ = \sec(60^\circ) = \frac{1}{\cos 60^\circ} = \frac{1}{1/2} = 2$

iv. $cosec(-690^\circ)$

Sol: $\cos ec(-690^\circ) = -\cos ec(690^\circ) = -\cos ec(7\times 90^\circ + 60^\circ) = -(-\sec 60^\circ) = \sec 60^\circ = \frac{1}{\cos 60^\circ}\frac{1}{1/2} = 2$

v. $\cot(1320^\circ)$

Sol: $\cot(1320^\circ) = \cot(14\times 90^\circ + 60^\circ) = \cot 60^\circ = \frac{1}{\tan 60^\circ} = \frac{1}{\sqrt{3}}$

vi. $\cos(-240^\circ)$

Sol: $\cos(-240^\circ) = \cos(240^\circ) = \cos(2\times 90^\circ + 60^\circ) = -\cos(60^\circ) = -\frac{1}{2}$

Q2. Express each of the following as a trigonometric functions of an angle of positive degree measure of less then45°

i. $\cos(168^\circ)$

Sol: $\cos(168^\circ) = \cos(180-12)^\circ = \cos(2\times 90-12)^\circ = -\cos(12^\circ)$

ii. $\sin(192^\circ)$

Sol: $\sin(192^\circ) = \sin(180+12)^\circ = \sin(2\times 90+12)^\circ = -\sin(12^\circ)$

iii. $\cos(333^\circ)$

Sol: $\cos(333^\circ) = \cos(360-27)^\circ = \cos(4\times 90-27)^\circ = \cos(27^\circ)$

iv. $\tan(213^\circ)$

Sol: $\tan(213^\circ) = \tan(180+33)^\circ = \tan(2\times 90+33)^\circ = \tan(33^\circ)$

v. $\cos(-435^\circ)$

Sol: $\cos(-435^\circ) = \cos(435^\circ) = \cos(5\times 90-15)^\circ = \sin(15^\circ)$

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vi. $\sin(219^\circ)$

Sol: $\sin(219^\circ) = \sin(180+39)^\circ = \sin(2\times 90+39)^\circ = -\sin(39^\circ)$

vii. $\tan(-597^\circ)$

Sol: $\tan(-597^\circ) = -\tan(-597^\circ) = -\tan(7\times 90-33)^\circ = -\cot(33^\circ)$

viii. $\cos(-111^\circ)$

Sol: $\cos(-111^\circ) = \cos(111^\circ) = \cos(90^\circ+21^\circ) = -\sin(21^\circ)$

ix. $\sin(-390^\circ)$

Sol: $\sin(-390^\circ) = -\sin(390^\circ) = -\sin(4\times 90^\circ+30^\circ) = -\sin 30^\circ$

Q3. Prove the following

i. $\text{Sin}(180^\circ+\alpha)\,\text{Sin}(90^\circ-\alpha) = -\text{Sin}\,\alpha\,\text{Cos}\,\alpha$

Sol. L.H.S = Sin (180°+ α) Sin 90° – α) = Sin (2 x 90° + α) Sin (1 x 90° – α) = (– sin α) (cos α)
= – Sin α Cos α = R.H.S

ii. $\sin 810^\circ \sin 630^\circ + \cos 135^\circ \sin 225^\circ = -1/2$

Sol: L.H.S $= \sin 810^\circ \sin 630^\circ + \cos 135^\circ \sin 225^\circ$

$= \sin(9\times 90^\circ+0^\circ)\sin(7\times 90^\circ+0^\circ)+\cos(90^\circ+45^\circ)\sin(180^\circ+45^\circ)$

$= (\cos 0^\circ)(-\cos 0^\circ)+(-\sin 45^\circ)(-\sin 45^\circ)$

$= (1)(-1)+\left(-\frac{1}{\sqrt{2}}\right)\left(-\frac{1}{\sqrt{2}}\right) = -1+\frac{1}{2} = -1/2 = \text{R.H.S}$

iii. $\tan 150^\circ \cot 330^\circ - 2\sec 135^\circ \, cosec 225^\circ = -3$

Sol: L.H.S $= \tan 150^\circ \cot 330^\circ - 2\sec 135^\circ \cos ec 225^\circ$

$= \tan(180-30)^\circ \cot(360-30)^\circ - 2\sec(180-45)^\circ \, cosec(180+45)^\circ$

$= \tan(2\times 90-30)^\circ \cot(4\times 90-30)^\circ - 2\sec(2\times 90-45)^\circ \, cosec(2\times 90+45)^\circ$

$= -\tan(30^\circ)(-\cot 30^\circ) - 2(-\sec 45^\circ)(-\cos ec 45^\circ)$

$= \left(-\frac{1}{\sqrt{3}}\right)(-\sqrt{3}) - 2(\sqrt{2})(\sqrt{2}) = 1-4 = -3 = R.H.S$

iv. $\sin 210^\circ + \cos 240^\circ + \tan 225^\circ + \cot 225^\circ = 1$

Sol: L.H.S $= \sin 210^\circ + \cos 240^\circ + \tan 225^\circ + \cot 225^\circ$

$= \sin(180+30)^\circ + \cos(180+60)^\circ + \tan(180+45)^\circ + \cot(180+45)^\circ$

$= \sin(2\times 90+30)^\circ + \cos(2\times 90+60)^\circ + \tan(2\times 90+45)^\circ + \cot(2\times 90+45)^\circ$

$= -\sin 30^\circ + (-\cos 60^\circ) + \tan 45^\circ + \cot 45^\circ$

$= -\frac{1}{2} - \frac{1}{2} + 1 + 1 = -1 + 1 + 1 = 1 = \text{R.H.S}$

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Q4. Prove that

i. $$\frac{\tan(180^\circ+\alpha)\cot(90^\circ-\alpha)}{\sin(360^\circ-\alpha)\cos(270^\circ+\alpha)}=-\sec^2\alpha$$

Sol: $$\text{L.H.S}=\frac{\tan(180^\circ+\alpha)\cot(90^\circ-\alpha)}{\sin(360^\circ-\alpha)\ \cos(270^\circ+\alpha)}=\frac{\tan(2\times90^\circ+\alpha)\cot(90^\circ-\alpha)}{\sin(4\times90^\circ-\alpha)\ \cos(3\times90^\circ+\alpha)}$$

$$=\frac{\tan\alpha.\tan\alpha}{(-\sin\alpha).\sin\alpha}=\frac{\tan^2\alpha}{-\sin^2\alpha}=\frac{\sin^2\alpha}{\cos^2\alpha}\times\frac{1}{-\sin^2\alpha}=-\frac{1}{\cos^2\alpha}=-\sec^2\alpha=\text{R.H.S}$$

ii. $$\frac{Sin^2(\pi+\theta)\tan(\frac{3\pi}{2}+\theta)}{Cot^2(\frac{3\pi}{2}-\theta)Cos^2(\pi-\theta)Cosec(2\pi-\theta)}=Cos\theta$$

Sol: $$\text{L.H.S}=\frac{Sin^2(\pi+\theta)\tan(\frac{3\pi}{2}+\theta)}{Cot^2(\frac{3\pi}{2}-\theta)Cos^2(\pi-\theta)Cosec(2\pi-\theta)}=\frac{(-Sin\theta)^2(-Cot\theta)}{\tan^2\theta(-Cos)^2(-Cosec\theta)}$$

$$=\frac{Sin^2\theta Cot\theta}{\frac{Sin^2\theta}{Cos^2\theta}Cos^2\theta\ \frac{1}{Sin\theta}}=Sin^2\theta\times\frac{Cos\theta}{Sin\theta}\times\frac{1}{Sin^2\theta}\times Sin\theta=Cos\theta=R.H.S$$

iii. $$\frac{Cos(90^\circ+\theta)\,Sec(-\theta)\tan(180^\circ-\theta)}{Sec(360^\circ-\theta)\ Sin(180^\circ+\theta)Cot(90^\circ-\theta)}=-1$$

Sol: $$\text{L.H.S}=\frac{Cos(90^\circ+\theta)Sec(-\theta)\tan(180^\circ-\theta)}{Sec(360^\circ-\theta)\ Sin(180^\circ+\theta)Cot(90^\circ-\theta)}=\frac{-Sin\theta\,Sec\theta(-\tan\theta)}{Sec\theta(-Sin\theta)\tan\theta}=\frac{Sin\theta\,Sec\theta\,\tan\theta}{-Sin\theta\,Sec\theta\,\tan\theta}$$

$$=-1=\text{R.H.S}$$

Q5. Show that $\sec\left(\frac{3\pi}{2}-\theta\right)\sec\left(\frac{5\pi}{2}-\theta\right)-\tan\left(\frac{3\pi}{2}-\theta\right)\tan\left(\frac{5\pi}{2}+\theta\right)=-1$

Sol: $$\text{L.H.S}=\sec\left(\frac{3\pi}{2}-\theta\right)\sec\left(\frac{5\pi}{2}-\theta\right)-\tan\left(\frac{3\pi}{2}-\theta\right)\tan\left(\frac{5\pi}{2}+\theta\right)$$

$$=(-cosec\,\theta)(cosec\,\theta)-(\cot\theta)(-\cot\theta)=cosec^2\theta+\cot^2\theta=-cosec^2\theta-1+cosec^2\theta=-1$$

Q6. If α,β,γ are angle of Triangle ABC, then prove that

i. $\text{Sin}(\alpha+\beta)=\text{Sin}\,\gamma$

Sol: As $\alpha+\beta+\gamma=180^\circ$ (sum of angles of triangle=180°)

$\alpha+\beta=180^\circ-\gamma$

$\text{Sin}(\alpha+\beta)=\sin(2\times90^\circ-\gamma)$

$\text{Sin}(\alpha+\beta)=\text{Sin}\,\gamma$ Hence proved.

ii. $$\sec\left(\frac{\alpha+\beta}{2}\right)=cosec\frac{\gamma}{2}$$

Sol: $\alpha+\beta+\gamma=180^\circ$

$$\alpha+\beta=180^\circ-\gamma\Rightarrow\frac{\alpha+\beta}{2}=\frac{180^\circ-\gamma}{2}\Rightarrow\frac{\alpha+\beta}{2}=90^\circ-\frac{\gamma}{2}$$

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$$\Rightarrow \sec\left(\frac{\alpha+\beta}{2}\right)=\sec\left(90^\circ-\frac{\gamma}{2}\right)$$

$$\sec\left(\frac{\alpha+\beta}{2}\right)=cosec\,\frac{\gamma}{2}$$

iii. $$cosec\alpha=\frac{1}{\sin(\beta+\gamma)}$$

Sol: $$\alpha+\beta+\gamma=180^\circ$$

$$\beta+\gamma=180^\circ-\alpha$$

$$\sin(\beta+\gamma)=\sin(180^\circ-\alpha)=\sin(2\times90^\circ-\alpha)$$

$$\sin(\beta+\gamma)=\sin\alpha$$

$$\sin(\beta+\gamma)=\frac{1}{cosec\alpha} \quad \text{or} \quad cosec\alpha=\frac{1}{\sin(\beta+\gamma)}$$

iv. **$\tan(\alpha+\beta)+\tan\gamma=0$**

Sol. let $\alpha+\beta+\gamma=180^\circ$

$\alpha+\beta=180^\circ-\gamma$

$\tan(\alpha+\beta)=\tan(2\times90^\circ-\gamma)$

$\tan(\alpha+\beta)=-\tan\gamma \Rightarrow \tan(\alpha+\beta)+\tan\gamma=0$ Hence Proved

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